Electroacoustics – loudspeakers and microphones

Electroacoustics is the study of devices that convert electrical signals into sound and vice versa: loudspeakers, microphones, headphones and hearing aids. Its great trick is to draw mechanical and acoustic systems as electrical circuits – then the whole transducer can be analysed with circuit theory. At DTH there were two courses, a basic and an advanced one. The page has an interactive loudspeaker box where you can compare closed and vented designs.

> DTH, courses 5121 and 5122 (1987), 6 + 6 points: Electroacoustics and Advanced electroacoustics. Denmark was – and is – a major power in electroacoustics: Peter L. Jensen co-invented the moving-coil loudspeaker with Edwin Pridham in California in 1915, and Brüel & Kjær, Bang & Olufsen, Jamo, Dynaudio, Scan-Speak, Vifa and Peerless were all close to DTH, which supplied many of their engineers.

The analogies

A vibrating mass on a spring with damping obeys the same differential equation as an electrical RLC circuit. This is exploited by translating mechanical and acoustic quantities into electrical ones. In the impedance analogy force corresponds to voltage and velocity to current:

Electrical	Mechanical	Acoustic
Voltage u	Force F	Sound pressure p
Current i	Velocity v	Volume velocity $q = Sv$
Inductance L	Mass m	Acoustic mass $\rho l / S$ (air in a tube)
Capacitance C	Compliance $C_m = 1/k$	Acoustic compliance $V/\rho c^2$ (air in a box)
Resistance R	Mechanical resistance R_m	Acoustic resistance (cloth, porous material)

A Helmholtz resonator is then an LC circuit, a loudspeaker box a capacitor, and a bass-reflex port an inductor. The whole chain from amplifier via voice coil, cone and box to the air outside can be drawn as one circuit and solved with the methods of circuit theory.

Transducer principles

Electrodynamic

A coil in a radial magnetic field. Current in the coil gives the force $F = Bl\,i$; if the coil moves, the voltage $u = Bl\,v$ is induced. The same factor Bl (the force factor) works both ways. Used in almost all loudspeakers and in dynamic microphones.

Electrostatic

A thin membrane in front of a fixed electrode forms a capacitor. If the distance changes, the capacitance changes, and with a polarising voltage this gives a signal. Used in condenser microphones and in electrostatic loudspeakers and headphones.

Piezoelectric

Certain crystals and ceramics (quartz, PZT) become electrically polarised when deformed. Stiff and suited to high frequencies – hence their use in ultrasound, tweeters and hydrophones.

Reciprocity

For linear, passive transducers a reciprocity principle holds: the transfer from electrical input to acoustic output is tied to the reverse path. It is used to calibrate measurement microphones absolutely – three microphones are measured in pairs against each other, and none of them needs to be known in advance.

The loudspeaker

In 1925 Chester Rice and Edward Kellogg of General Electric described the loudspeaker we still use: a light cone, suspended so that it can move freely as a piston, driven by a voice coil. At low frequencies the cone is small compared with the wavelength and radiates as a monopole. The far-field sound pressure is proportional to the cone's *acceleration*. Above the cone resonance the motion is mass-controlled: the acceleration is constant, and the frequency response is flat. That is why a loudspeaker can cover several octaves.

Without a box the sound from the front and back short-circuits at low frequencies: they are in antiphase and cancel. An infinite baffle – a wall – solves the problem but is impractical. Instead the

rear is enclosed in a box.

The Thiele–Small parameters

The Australian engineer Neville Thiele (1961) and Richard Small (1972–73) showed that the low-frequency behaviour of a loudspeaker driver is fully determined by three quantities that can be measured from the electrical impedance curve:

- f_s – the free-air resonance frequency, set by the mass of the cone and the stiffness of the suspension.
- Q_{ts} – the total quality factor, made up of a mechanical and an electrical part: $1/Q_{ts} = 1/Q_{ms} + 1/Q_{es}$. The electrical damping arises because the cone motion induces a back-EMF that the amplifier short-circuits.
- V_{as} – the volume of air that has the same stiffness as the suspension.

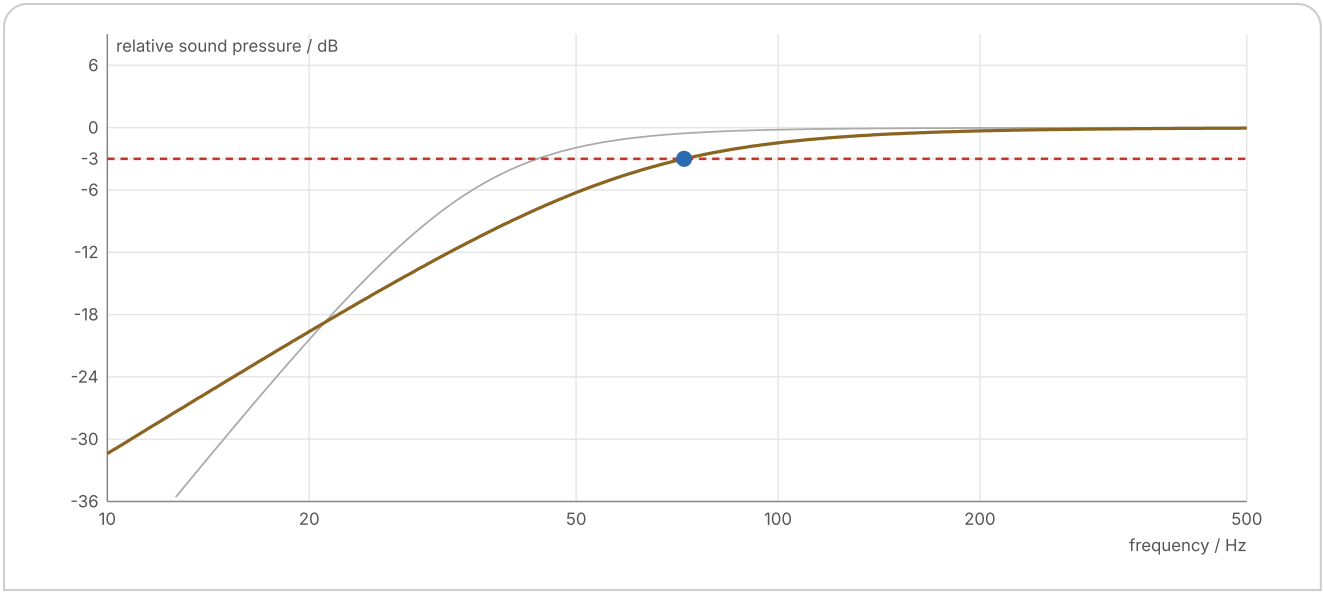
Thiele's great insight was that driver and box together form a high-pass filter – second order for a closed box, fourth order for a bass reflex – and that the box dimensions can be chosen to give the filter a known shape, such as Butterworth (maximally flat).

Closed and vented boxes

In a closed box of volume V_b the air acts as an extra spring. With the compliance ratio $\alpha = V_{as}/V_b$ the system resonance and quality factor become

$$f_c = f_s\sqrt{1+\alpha}, \qquad Q_{tc} = Q_{ts}\sqrt{1+\alpha}, \qquad H(s) = \frac{s^2}{s^2 + s\,\omega_c/Q_{tc} + \omega_c^2}$$

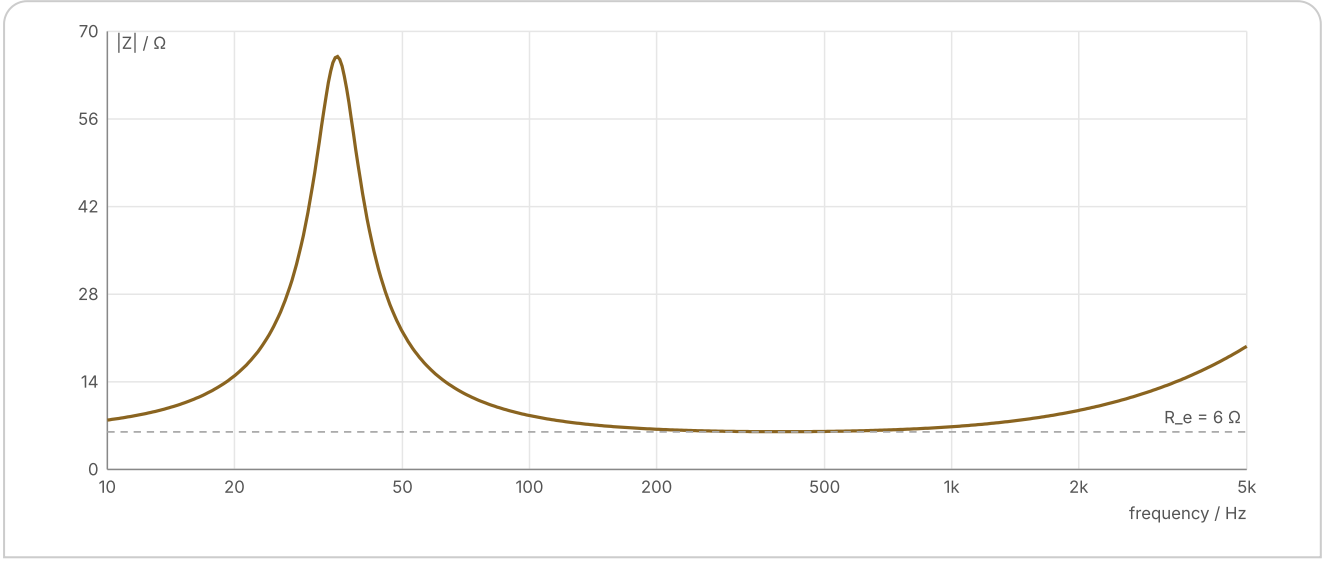
$Q_{tc} = 0.707$ gives the flat Butterworth curve; a higher Q_{tc} gives a hump and "boomy" bass. In a bass reflex the box has a tube (port) whose air mass resonates with the air spring of the box at f_b. Around f_b the cone hardly moves, and the sound comes from the port. The response falls by 24 dB per octave instead of 12, but can go deeper with the same box and driver.



The impedance curve

Seen from the amplifier the loudspeaker is a resistance R_e (the DC resistance of the voice coil) and a small inductance in series with a parallel resonant circuit, which is the mechanical system of the cone translated through Bl: the mass becomes a capacitor, the compliance of the suspension an inductor. At resonance the impedance has a peak. From its height and width Q_{ms} and Q_{es} can be found – and with an extra measurement with a known mass on the cone or in a known box, also V_{as}. That is how Thiele–Small parameters are measured in practice.

$$Z(f) = R_e + j\omega L_e + \frac{R_{es}}{1 + jQ_{ms}\left(\frac{f}{f_s} - \frac{f_s}{f}\right)}, \qquad R_{es} = R_e \frac{Q_{ms}}{Q_{es}}$$



Microphones

The condenser microphone was invented by E. C. Wente at Bell Labs in 1916, and Brüel & Kjær, founded by Per Brüel and Viggo Kjær in 1942, began building measurement microphones. Their half-inch condenser microphone, with a diaphragm of a few micrometres of nickel, became the world standard for acoustic measurements. The diaphragm must be stiff enough for its resonance to lie above the useful frequency range, and the cavity behind it is damped so that the response is flat. The microphone is pressure-controlled: the diaphragm moves in proportion to the sound pressure.

At high frequencies the microphone itself disturbs the sound field because it is no longer small compared with the wavelength: the pressure on the diaphragm rises at frontal incidence. That is why there are free-field, pressure and diffuse-field microphones, each corrected for its measurement task. To measure sound intensity, two microphones are placed face to face with a spacer – the method developed at DTH and B&K around 1980.

Since 1989

- **DSP and active loudspeakers:** Digital filters handle crossovers and correction of room and driver. Bang & Olufsen's BeoLab 5 (2003) measures the room itself and adapts the bass.
- **Neodymium magnets** (from the mid-1980s) have made loudspeakers and headphones light and small.
- **MEMS microphones** of silicon, less than a millimetre across, now sit in billions of phones, hearing aids and laptops.
- **Class-D amplifiers** with efficiencies above 90 % have replaced the heavy analogue amplifiers – ICEpower was developed at B&O based on research at DTU.
- **Simulation:** Whole drivers are now simulated with coupled FEM models of magnet, mechanics and acoustics.

Further reading

- Leo L. Beranek and Tim Mellow: *Acoustics: Sound Fields, Transducers and Vibration* (2nd ed., Academic Press 2019)
- A. N. Thiele: "Loudspeakers in vented boxes", *Proc. IRE Australia* 1961, reprinted in *J. Audio Eng. Soc.* 1971
- Richard H. Small: "Closed-box loudspeaker systems" and "Vented-box loudspeaker systems", *J. Audio Eng. Soc.* 1972–73
- Finn Jacobsen and Peter Juhl: *Fundamentals of General Linear Acoustics* (Wiley 2013)